

Sugarcane (*Saccharum officinarum*) – Reference Guide

Overview & Classification

Sugarcane, commonly called *sugar cane*, is a tall perennial grass in the genus *Saccharum* (family Poaceae) [22†L82-L87] . The principal cultivated species is *Saccharum officinarum* L. (often grown as interspecific hybrids). It is native to New Guinea (Pacific region of Asia) and was domesticated there [22†L90-L91] [24†L77-L80] . (The genus is in tribe Andropogoneae of the grass family.) Sugarcane is grown worldwide in tropical/subtropical zones, with major production in India, Brazil and Southeast Asia.

Climate & Growing Conditions

Sugarcane thrives in tropical to warm subtropical climates. It needs a **long, frost-free warm season** with high light and moisture [10†L76-L84] . Optimum mean temperatures are about **22–30 °C**, with germination best at 32–38 °C [10†L82-L86] [26†L222-L226] . It can survive short spells as high as ~42 °C and as low as ~15 °C [8†L4-L8] , but growth slows sharply below 20 °C and stops below ~12 °C [26†L222-L226] [46†L38-L40] . Bright sunshine (7–9 hours/day) and high humidity (40–85% relative humidity) favor good vegetative growth [26†L250-L258] . A mild cool, dry period before harvest helps ripening of sugars.

Rainfall must be abundant: **1,500–3,000 mm per season** is ideal in the tropics [17†L527-L531] [26†L232-L237] . (Subtropical cane often uses irrigation to provide ~1,400–1,500 mm/yr [53†L1752-L1759] [17†L527-L531] .) Sugarcane can be irrigated during dry spells; flood/furrow or drip irrigation are used. The crop is sensitive to waterlogging and requires well-drained conditions.

A single sugarcane crop takes **12–18 months** from planting to maturity (plant crop), with ratoon (regrowth) crops ~10–12 months each [10†L89-L92] . In research and plantation conditions, plant-cane durations of 15–16 months are common [10†L89-L92] . In practice, planting and harvest seasons vary: for example, India uses spring (Feb–Mar) and autumn (Nov–Dec) plantings depending on climate. Regions between about 35°N and 35°S latitude are suitable [10†L76-L84] . In India, Uttar Pradesh, Maharashtra and Karnataka dominate production (together ~80% of national cane) [26†L189-L194] .

Soil Requirements

Sugarcane grows best on **deep, fertile, well-drained loams or clay-loam soils**. Ideal soils are >1 m deep (roots can reach 3–4 m) with good aeration and moisture-holding capacity [10†L100-L107] . Avoid waterlogged or saline soils: high water tables or standing water encourage root diseases (e.g. red rot) [10†L100-L107] [17†L536-L540] . Optimum soil pH is near neutral (~6.0–7.5); can tolerate a range ~5.0–8.5 [10†L100-L107] .

Soil preparation: Deeply plough or disc the field (30–40 cm) to break hard pans and incorporate organic matter. Ridges or furrows are often formed for planting. A basal application of well-rotted

manure/compost (10–15 t/ha) or pressmud (5–8 t/ha) is typically added before final ploughing [56†L1921-L1924] . This improves soil structure and fertility. Gypsum or lime can be applied if needed to correct pH. After ploughing, harrow to a fine tilth and level the field.

Propagation & Sowing

Sugarcane is propagated **vegetatively** from stem cuttings (“setts”), not from true seed. Healthy, disease-free setts (usually 2–3 buds each) are used. Seed rate is high: typically **6–10 tonnes of setts per hectare** (about 20,000–40,000 two-budded setts/ha) [46†L28-L36] . In India, seed rate and planting density vary by zone: e.g. ~25,000–35,000 setts/ha in peninsular areas but ~50,000–75,000/ha in cooler north (since germination may be poorer there) [59†L1472-L1480] . Two- or three-budded setts give stronger stools than single buds [46†L28-L36] . Use setts ~8–10 months old from the previous cane harvest.

Planting is usually in flat furrows or on ridges. Cover each sett with about **5–7 cm of soil** (enough to keep it moist but allow sprouting) [46†L44-L47] . Spacing: common row spacing is about **75–100 cm**; within-row spacing ~20–30 cm. For example, plant-cane may be set at 90 cm × 20 cm (~55,000 plants/ha) or in paired rows (75+75 cm) to facilitate drainage [42†L751-L758] [59†L1466-L1474] . (One study found 75–90 cm row spacing gave highest yields [42†L751-L758] .)

Sowing season depends on region. In tropical India, sugarcane may be planted almost year-round with irrigation. In subtropical areas it is timed after rains (spring or autumn). Sprouting is fairly slow: buds germinate in **10–20 days** under warm (26–32 °C) moist soil [46†L38-L40] . Under cool or dry conditions germination can fail. Gap-filling (re-planting missing stools) should be done within 30 days to maintain stand uniformity [40†L1-L4] [46†L39-L47] .

Irrigation Management

Sugarcane is a **water-intensive** crop. Total water need is about 1,500–3,000 mm over the crop cycle (more in tropical regions) [17†L527-L531] [26†L232-L237] . In practice, irrigation is managed by growth stage:

- **Germination (0–35 days):** Maintain uniformly moist soil. Irrigate every 7–10 days (2–3 times) to ensure bud sprouting [26†L239-L246] .
- **Tillering (35–100 days):** Cane produces most tillers; water every 8–12 days depending on weather and soil. Adequate moisture now supports stool development.
- **Grand-growth (100–270 days):** This vegetative phase (stem elongation) has the highest water demand. Irrigate roughly once per week (about 6–8 irrigations) [26†L239-L246] .
- **Ripening (270 days to harvest):** As the cane matures, reduce watering. Irrigation can be cut back or stopped ~1–2 months before harvest to concentrate sugars [26†L239-L246] . (Minimal irrigation is needed once ripening begins.)

Critical phases are early establishment and the grand-growth phase; moisture stress during these reduces yield. Conversely, excess water at harvest can dilute sugars or cause lodging. Overall irrigation methods may include furrow/flood, sprinkler or drip. Modern systems use drip/micro-irrigation to save water. (For drip: lay laterals in furrows ~20 cm deep, with setts ~5 cm above the dripper line [5†L438-L442] .)

Soil moisture should be kept near field capacity during active growth. Subtropical cane (shorter crop) often relies partly on stored soil moisture. In any region, high irrigation efficiency is desirable (surface irrigation efficiencies in sugarcane are often <50% [53†L1752-L1759]).

Nutrient & Fertilizer Management

Sugarcane is a **heavy feeder** crop. Nutrient requirements are high, especially nitrogen (N) and potassium (K), with moderate phosphorus (P). Approximate nutrient needs to produce ~100 t cane/ha are on the order of **120–200 kg N, 20–80 kg P₂O₅, and 125–160 kg K₂O/ha** [10†L109-L113] . Indian recommendations vary by zone (plant crop): e.g.

- North-Western India: ~150 kg N, 60 kg P₂O₅, 60 kg K₂O/ha [58†L1979-L1987] .
- Peninsular/East Coast India: ~250 kg N, 100 kg P₂O₅, 125 kg K₂O/ha [58†L1979-L1987] .
(In ratoon crops, N rates are often ~25% higher – e.g. 200–250 kg N/ha – because stubble requires more N for regrowth [58†L1979-L1987] [54†L49-L53] .)

Application: Apply most phosphorus and K as a basal (pre-plant) dose. Nitrogen is split: for example, half at planting and half at ~60–90 days after planting [59†L1458-L1460] . (In practice, common schedules are 2–3 splits of N: at planting, at crown development, and at grand-growth phase [59†L1458-L1460] [54†L49-L53] .) For ratoons, split N into 3 equal doses (Feb–Mar, April, May) [54†L49-L53] .

Micronutrients may be needed on some soils: **Sulfur (S)** is often applied (40–60 kg/ha) in S-deficient areas, typically as gypsum [54†L25-L28] . Zinc or boron are used if soil tests indicate deficiency.

Organic manures: Farmyard manure or compost (10–15 t/ha) is recommended at planting to improve soil fertility [56†L1921-L1924] . Mulching with cane trash or pressmud (sugar factory residue) can also supply nutrients and conserve moisture.

Varietal differences matter: high-yielding hybrid varieties (often grown in the south) can require N at the upper end of the above ranges, whereas subtropical varieties may use somewhat less. Soil-testing and local guidelines should refine these rates.

Growth Stages

Sugarcane development can be divided into stages from planting (germination) to maturity. Typical durations (after planting) are:

- **Germination/Establishment (0–30 days):** Buds sprout into shoots. A few leaves emerge; “nap” can appear at the cane base. Seed piece establishes.
- **Tillering (30–90 days):** The plant stool produces side shoots (tillers). Leaf area and tiller number rapidly increase. This stage determines the number of cane stalks per stool.
- **Grand Growth/Elongation (90–270 days):** Maximum leaf area and stem elongation occur. Internodes lengthen and thicken, accumulating biomass. This vegetative growth phase is the longest (often 3–9 months) and accounts for most cane and juice yield.
- **Maturity/Ripening (270 days onward):** Growth slows, and sugars accumulate in the stalk. Lower leaves yellow and die as nutrients translocate to the cane. Cane begins to ripen; brix (soluble solids) rises. Total crop duration is generally 12–18 months (≈360–540 days) for plant cane [10†L89-L92] . Ratoon crops cycle faster (≈10–12 months).

Sugarcane is normally harvested at or near full maturity (just before excessive leaf fall). Flowering is usually prevented by short-day selection, since flowering can reduce sugar content 【10†L93-L99】 .

Pest Management

Key insect pests include stem borers, sap-suckers and soil insects. Common pests are: Early shoot borer (*Chilo infuscatellus*, also called “dead heart borer”), Internode (stem) borers, Top borer (*Scirpophaga nivella*), whitefly, sugarcane woolly aphid (*Ceratovacuna lanigera*), mealybug, white grub (larvae of *Holotrichia* beetles), termites, and grasshoppers 【61†L15-L20】 . For example, the early shoot borer damages the growing point, causing “dead hearts” in young cane (especially in late-planted crops) 【68†L536-L540】 . Soil pests like white grubs feed on roots and can kill young stools. Sap-suckers (aphids, mealybug, whitefly) cause yellowing and sooty mold.

Management combines cultural, biological and chemical methods (IPM). Cultural controls: use clean seed (disease- and pest-free setts), proper field sanitation and ploughing (e.g. summer ploughing to expose grubs/termites), timely planting, and crop rotation. Remove and destroy infested stools or trash to break pest cycles. Biological controls: releases of egg/larval parasitoids (e.g. *Trichogramma chilonis* for shoot borer eggs, *Cotesia* wasps for borers) and predators (e.g. ladybird beetles for mealybug, syrphid flies for aphids) are used. *Bacillus thuringiensis* (Bt) products and entomopathogenic fungi (e.g. *Metarhizium* for soil pests) can suppress borers and grubs.

Chemical control is used as needed: e.g. granular or spray insecticides at planting against borers and grubs (such as chlorantraniliprole, fipronil or imidacloprid applied in furrow), and foliar sprays (e.g. for aphids or top borer). Sprays of systemic insecticides can protect against sucking pests, and granular carbofuran or chlorpyrifos at planting can reduce termites/white grub attack. Insecticide choice and timing depend on monitoring: for instance, scout fields 10–20 days after planting for borer holes, and spray if damage exceeds economic thresholds.

(As an example, *Trichogramma* wasps are often released at 10-day intervals to parasitize borer eggs, and sticky traps or pheromone traps help monitor adult moth flights. Intercropping or trap crops may also be used to reduce pest pressure.)

Disease Management

Major diseases of sugarcane in India include **red rot**, **grassy shoot**, **pokkah boeng**, **smut**, **leaf scald**, **mosaic**, and bacterial wilt. Key examples:

- **Red rot** (*Colletotrichum falcatum*): The most destructive disease; causes white pustules and a sour smell inside cut cane, with hollowing of the stalk. Control by using disease-free setts, **hot-water treatment** of seedcane (52 °C for 10–15 minutes) and resistant cultivars 【72†L79-L84】 . Foliage sprays of systemic fungicides or Bordeaux paste on setts can help.
- **Smut (whip smut)** (*Ustilago scitaminea*): Causes black “whip” (teliospore) protrusions from the growing shoot, plus stunting and leaf bunching. Control by clean planting material, roguing of infected plants, and avoiding ratoon cropping 【72†L109-L112】 .
- **Grassy shoot** (a phytoplasma): Leads to proliferation of slender, leaflike shoots (no cane formation). It is insect-transmitted (by leafhoppers). Management includes hot-water treatment of setts (52 °C, 30 min) and applying oxytetracycline injections in severe cases; also removing infected stools.
- **Pokkah boeng** (*Fusarium* sp. or *Phoma*): Causes twisted, chlorotic or necrotic young leaves and stalk tip rot. Reduced by resistant varieties and fungicide treatment of seedcane.

- **Leaf scald** (*Xanthomonas albilineans*): Bacterial streaks on leaves and white striping in cane. Use of clean seed and weed control (eliminating grass weeds which harbor the bacterium) is advised.
- **Mosaic, Yellow-leaf (SCYLV), and Yellow-Leaf (Sugarcane yellow leaf virus)**: Viral diseases causing yellow patterns on leaves and reduced growth. Control by controlling aphid/planthopper vectors and using virus-free planting material.

In general, disease management relies on **healthy seedcane and resistant varieties**. For instance, variety Co 0238 is moderately resistant to prevalent red rot races 【81†L99-L105】. Hot-water treating setts and field sanitation (destroying diseased clumps) are important. Fungicides (e.g. carbendazim, propiconazole) can be applied to setts or foliage for fungal diseases, following local guidelines. Crop rotations and intercropping (e.g. with legumes) can also reduce disease carry-over.

Weed Management

Weeds compete strongly with young cane. The **critical weed-free period** is the first 2–3 months after planting, when the crop canopy is developing. Early and effective weed control is essential. A common strategy is a combination of herbicide plus manual weeding:

- **Pre-emergence herbicide**: Atrazine at 1.0–2.0 kg/ha applied at or shortly after planting controls grasses and broadleaf weeds 【75†L487-L493】. (Oxyfluorfen is another option.) Apply through a sprayer at 3–7 days after planting. 【75†L487-L493】.
- **Pre-plant herbicide**: Glyphosate (2.0 kg/ha) sprayed 2–3 weeks before planting can kill existing perennial weeds (e.g. nutgrass) 【75†L500-L504】. At planting, a shallow rotary hoeing or pre-emergence herbicide ensures a clean seedbed.
- **Post-emergence control**: Hand-weeding or mechanical weeding along the row at ~30, 60 and 90 days after planting removes escapes 【75†L487-L493】. (In fields with high weed pressure, additional hoeing at 20–25, 45–50 days is recommended 【74†L19-L22】.) For broadleaf weeds (e.g. **Striga**), a follow-up spray of 2,4-D is effective 【75†L493-L499】. For nutgrass (*Cyperus*), spot-treat with flupyr-sulfuron or similar.
- **Intercropping**: Growing legumes (e.g. soybean, groundnut) between cane rows in the first months can suppress weeds and fix N.

Overall, keep the field weed-free for at least the first 60–90 days. Well-timed herbicide (Atrazine) plus 2–3 hand hoeings is the usual practice 【75†L487-L493】 【75†L500-L504】. This ensures the fast-growing cane (leaf area index ↑) can shade out later weeds.

Harvesting

Maturity indicators for cane include drying/loss of lower leaves, blackening of the internode joints, and high juice Brix at the bottom of the stalk. (A simple field test is when refractometer readings (“Brix”) of juice from top and bottom of stalk are similar.) In practice, harvesting is timed by age (e.g. 10–14 months in the tropics; 12–16 months in subtropics) and by regional schedule.

Harvesting is usually **manual** (cutting the stalk above ground with a machete) or by mechanical harvesters where available. Most cane is harvested “green” (without burning trash) to preserve soil organic matter; however, stalk-burning is done in some areas to remove leaves before cut. After harvest, older stump is left (~2 cm) to ratoon.

The cut cane is transported to mills or processing units as soon as possible (ideally within 24–48 hours) since sucrose degrades over time. Delays in crushing can reduce recoverable sugar by ~0.2–0.3% per day. Prompt transport on rail or road trucks is common.

Post-Harvest Handling

Sugarcane has **very limited shelf life**. It should be processed soon after cutting because sucrose quickly inverts and ferments in the stalk. Generally, mills start crushing the day of or next day after harvest. If delays occur (e.g. for field-windrowed cane), shade and ventilation (or minimal irrigation) are used to reduce heat build-up. Cane can be stored intact for 1–2 days with only a few percent loss in sugar.

After crushing, raw juice is clarified and concentrated to make sugar or fermented for ethanol. Bagasse (fibrous residue) is dried and used as fuel for cogeneration of steam/electricity at the mill. Other by-products include pressmud (filter cake) used as fertilizer and molasses used for alcohol/jaggery production. Regions may process cane into jaggery (khandasari) or dried cubes, depending on infrastructure.

Yield & Economics

Average sugarcane yields vary widely. In India, typical yields are **60–80 t/ha** (plant cane + first ratoon) under good management. For example, national data (2015–16) report ~5.0 million ha producing 356 million t cane (≈ 71 t/ha) [17†L509-L513] . High-potential irrigated farms can reach 100–150 t/ha with improved hybrids. World averages are around 75–90 t/ha.

Factors affecting yield include variety, climate, irrigation, nutrients, and management. Adequate water, balanced fertilization, pest/disease control and good cultivation boost yields. For example, planting a high-yielding variety like Co 0238 increased sugarcane yields by ~20% in North India [81†L99-L105] .

Economics: Sugarcane is India's key cash crop. Income depends on cane productivity and sugar price. High inputs (water, fertilizers) raise costs, but waste biomass (bagasse) provides energy credit. Regional payments to farmers vary by state policy. As an illustration, increased use of Co 0238 in UP raised cane production and mill sugar recovery, giving farmers several thousand rupees extra profit per hectare [81†L118-L126] . (Exact figures depend on local prices and subsidies.)

Common Varieties/Cultivars

Hundreds of sugarcane cultivars are used worldwide. In India, regionally adapted varieties are named by institutes (e.g. "Co", "CoC", "CoS", "CoJ" for Coimbatore; "CP" for Canal Point, Florida; "HoCP" for Houma, Louisiana). Key Indian varieties include:

- **Co 86032 ("Nayana")** – a mid-late maturing variety developed by ICAR-SBI; very high yielding (~150–160 t/ha under trials) with high sugar content and good ratooning. It shows moderate resistance to smut [78†L9-L15] .
- **Co 0238 (Karan 4)** – an early-maturing high-sugar variety released for North-West India (Punjab, Haryana, UP). It yields ~80–85 t/ha of cane and significantly outperformed older checks [81†L99-L105] ; moderately resistant to red rot. It has become widely adopted in Punjab and UP [81†L118-L126] .
- **Co 87261 (Tulsiman)** – a popular variety in Maharashtra; high sugar, medium maturity.

- **Co 11015** – a high-yielding, high-sugar variety used in tropical regions (Maharashtra).
- **Co 7717** – an early maturing variety for North Zone.
- **HoCP 85-845** – high-sugar Floridian variety used in parts of India.

There are also specialty varieties (e.g. “purple” sugarcane, chewing cane) grown in small areas. Breeding priorities include higher sugar yield, stress tolerance and pest/disease resistance. Farmers choose varieties based on regional recommendations and crop schedule.

Regional Considerations

Sugarcane practices vary by climate zone:

- **Subtropical North (e.g. Punjab, Haryana, UP):** two main crops (spring and autumn), often irrigated year-round. Growing season ~10–12 months. Annual water need ~1,400–1,500 mm [53†L1752-L1759] . Fertilizer rates are moderate (e.g. 150–180 kg N/ha plus 60–80 P and 60 K) [56†L1916-L1921] . Short-duration, cold-tolerant, early-maturing varieties (e.g. Co 0238, Co J 64) are used. Pest/disease pressures include borers and woolly aphid in cooler weather.
- **Tropical Peninsular (e.g. Maharashtra, Karnataka, TN):** usually one long season (planting around Nov–Dec, harvesting Oct–Nov), with ratooning. Season ~15–18 months. Rainfall is heavier (monsoon 800–1200 mm) plus irrigation up to 2,000–2,500 mm total. Fertilizer needs are higher (often 250–350 kg N/ha, 100 P, 100–125 K) [56†L1916-L1921] [58†L1979-L1987] . Tropical, high-sugar hybrids (e.g. Co 86032, Co 7317) are common. Pests like pyrilla (leaf hopper) and borers may be more prevalent in humid tropics.
- **Eastern India and Sub-Himalayan (e.g. Bihar, UP East, Uttarakhand):** intermediate climate; crop duration ~12–15 months. Moisture 1500+ mm. Medium N (150–200 kg). Varieties from either tropical or subtropical pools may be grown.
- **Climate extremes:** In high altitude or dry regions, sugarcane grows poorly or requires greenhouse conditions.

In each region, soil type and irrigation methods also dictate practice (e.g. sandy alluvial soils in Punjab vs. lateritic red soils in Maharashtra). Adaptation measures (variety choice, planting date, irrigation scheduling) are made accordingly.

Sources: Authoritative agronomy and extension literature on sugarcane cultivation [22†L82-L87]

[10†L82-L90] [26†L222-L226] [56†L1916-L1921] [58†L1979-L1987] [68†L536-L540] [72†L79-L84] [75†L487-L493] [78†L9-L15] [81†L99-L105] .
